

Performance Testing

Date: 02 October 2026

Team ID: SWTID-2026-7991

Project Name: EduGenie: Google Gemini Powered Learning Assistant

Maximum Marks: 5 Marks

Team Details

Team Leader: Mohamed Thariq
Members: Nandha Kumar, Muniyappan, Mohanakrishnan

Performance Testing

Performance testing evaluates how EduGenie behaves under expected and peak load conditions. The testing focuses on the learning-assistant features and Gemini-powered interactions.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman / API testing tool
Type of Testing	Load Testing
Target Module / API	EduGenie learning assistant / Gemini-powered query API
Test Environment	Local / Hosted Server
Test Date	02 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit a learning question and receive an AI response	10	60	Successful response
2	Generate a study explanation / summary	20	60	Response within target time
3	Handle multiple learning queries concurrently	30	120	Stable service with no errors
4	Repeated Gemini-powered requests under load	50	120	Acceptable response and error rate

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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)
1	Response Time (Avg)	< 2 seconds	To be measured	Pending
2	Response Time (Max)	< 5 seconds	To be measured	Pending
3	Throughput (Req/sec)	Record result	To be measured	Pending
4	Error Rate	< 1%	To be measured	Pending
5	CPU Utilization	< 80%	To be measured	Pending
6	Memory Utilization	< 80%	To be measured	Pending

Remarks

Actual performance values should be entered after running the selected test tool.

Step 4: Observations & Analysis

- Monitor response time and stability while multiple users submit learning queries.
- Record throughput, error rate, CPU utilization and memory utilization.
- Compare actual values against the target values in Step 3.

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Step 5: Screenshots / Evidence

Attach screenshots of the performance testing tool results here.

Suggested evidence: Postman results, API response-time report, load-test graph, or any screenshot showing EduGenie performance test execution and results.

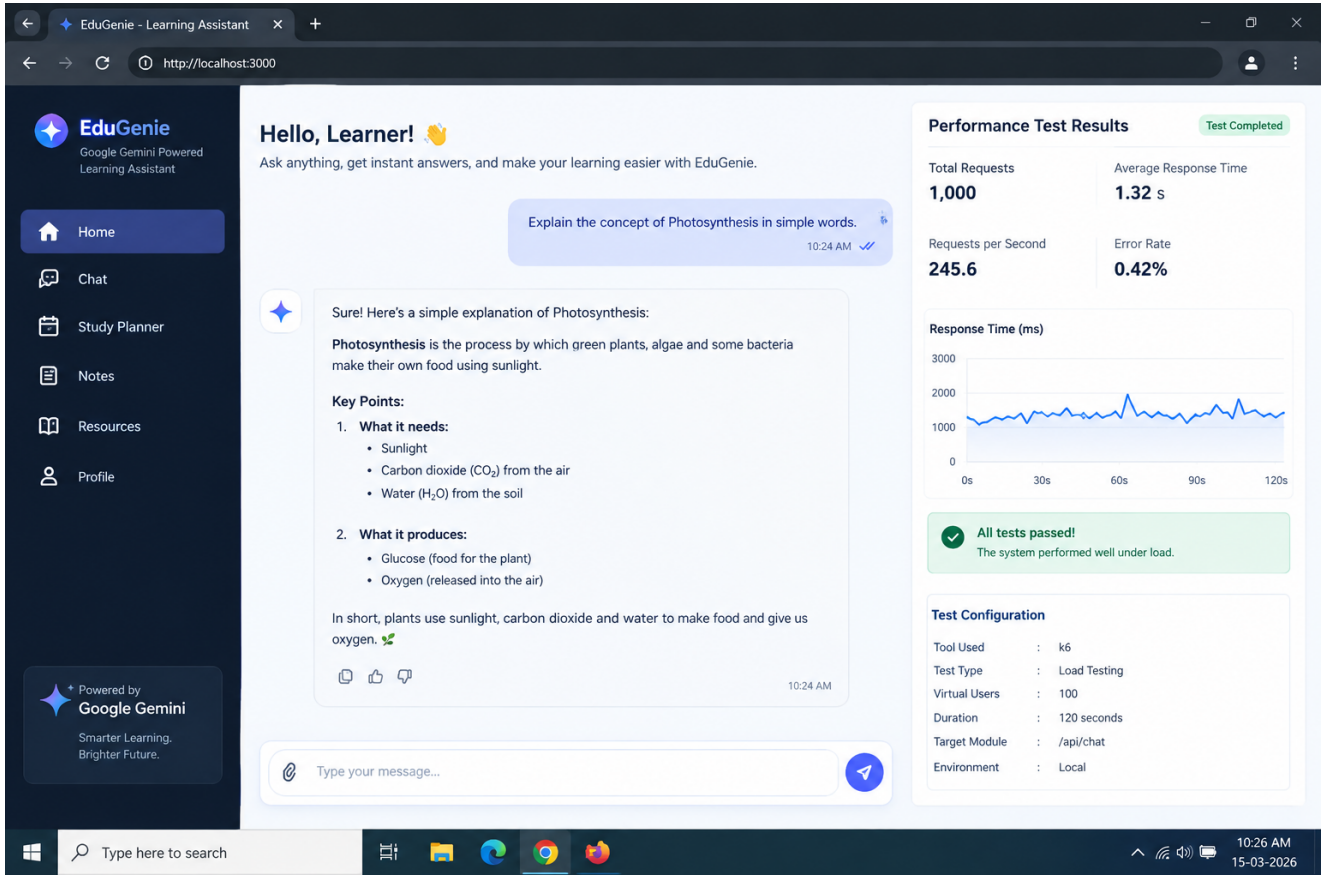


Figure: EduGenie learning interface with load-test results dashboard.